SHUJIA (IRIS) CHEN

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EDUCATION

B.S. in Data Science, Minor in Cognitive Science Sep 2024 – Mar 2027

University of California, San Diego (UCSD), La Jolla, CA

Halıcıoğlu Data Science Institute

Coursework: Data Structures and Algorithms, Statistical Data Analysis and Inference, The Practice and Application of Data Science, Data Visualization, Stats Data Analysis, Web Mining and Recommender Systems

TECHNICAL PROJECT

Global Warming Visualization: Surface Air Temperature Mapping (2015–2030) Fall 2025

- Architected an interactive geospatial platform using **D3.js**, **TopoJSON**, and climate-model grid data to visualize global surface temperature trends and anomaly patterns from 2015–2030.
- Developed a scalable data pipeline to **ingest, decompress, and preprocess** multi-year temperature fields, enabling real-time map updates, anomaly mode switching, and time-series playback across 180+ frames.
- Delivered a responsive UI ecosystem with zoom/pan navigation, dynamic sliders, and region-level selection workflows, enhancing user engagement and supporting comparative analysis of warming trajectories.

Amazon Review Analysis/Rating Prediction Fall 2025

- Developed a predictive model to estimate user ratings (1-5 stars) based on Amazon product review text and metadata, aiming to validate the effectiveness of interpretable NLP baselines.
- Processed large-scale text corpora using **TF-IDF** vectorization and engineered features from review metadata; implemented and tuned a **Ridge Regression** model to capture rating correlations.
- Demonstrated that the TF-IDF + Ridge Regression approach significantly outperformed baseline models, establishing a highly efficient and interpretable baseline for rating prediction on massive datasets.

Bikewatching: Boston Bike Traffic Interactive Visualization Fall 2025

- Developed an interactive geospatial platform to analyze and visualize the spatial-temporal distribution of BlueBikes traffic across Boston and Cambridge.
- Leveraged **Mapbox GL JS** as the map framework, combined with **D3.js** and SVG overlays, to process **260,000+** ride records and implement dynamic data filtering via a custom time slider.
- Achieved real-time rendering for **260+** stations, utilizing dynamic circle sizing and color-coding to represent traffic volume and flow balance, significantly enhancing visual clarity and user engagement.

Dark Data: Exploring Power Outage Patterns Spring 2025

- Investigated the impact of climate conditions and residential electricity prices on outage duration/frequency using **hypothesis testing** and **regression modeling**; **published** results via an interactive webpage.
- Executed data cleaning and feature engineering, **developed** a **Random Forest Regression** model, and **applied** permutation testing to assess fairness.
- Reduced prediction error by **7%** (8,919 $\rightarrow$ 8,292 minutes) and enhanced result accessibility through **web-based visualization**, providing actionable insights for grid resilience policy.

TECHNICAL SKILLS

Programming Languages: Python, SQL, HTML/CSS, JavaScript, Java, R
Libraries: Pandas, NumPy, D3.js, JSon
Tools: Excel, Jupyter Notebook, Git, GitHub, VS code, PyCharm, Eclipse, SQLite, Blender, RStudio
Certifications: Project Management Essentials, Corporate Finance Fundamentals